

Intrusion detection using logistic regression: A comparison of regularization methods and PCA-logistic regression method

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Abstract

In this report, we examine the performance of four regularization techniques and PCA-logistic regression in an intrusion detection task. To be more precise, we transformed the intrusion challenge into a binary classification issue and examined the performances using various regularizations and logistic regression. It was discovered that No regularization picked a suitable amount of helpful features and offered a decent accuracy of 95.9% using a real-time IoT dataset. The best accuracy, 98.7%, was obtained using the Lasso regularization technique.

1 Introduction

The widespread adoption of networked systems in today's digitally connected world has enabled unprecedented breakthroughs in communication, trade, and cooperation. The integrity and confidentiality of sensitive data are seriously threatened by malicious activity and illegal access, which has resulted in a multitude of cybersecurity concerns as a result of this digital progress. As a proactive tool for spotting and averting possible security breaches, intrusion detection systems become essential for protecting digital assets. The challenging task of differentiating between malicious efforts to compromise system integrity and lawful network activities falls under the purview of intrusion detection. Modern attack vectors are too complex for conventional security measures to stop, as cyber dangers get more complex. Strong intrusion detection systems (IDS) are therefore essential for bolstering cyber defenses and maintaining the reliability of digital infrastructures. Furthermore, successful intrusions can have far-reaching consequences, including financial damage, harm to one's reputation, and even jeopardizing interests in national security. Therefore, safeguarding both human privacy and the integrity of vital digital ecosystems emphasizes the necessity of developing and implementing improved intrusion detection technologies. Researchers and industry professionals have developed an extensive array of techniques to deal with the complexities of intrusion detection throughout the years. The field of intrusion detection systems

(IDS) methodologies is broad and dynamic, ranging from signature-based strategies that depend on pre-established patterns to anomaly detection methods that examine departures from ordinary activity. Neural networks, support vector machines, and ensemble techniques are just a few of the machine learning algorithms that have shown promise in identifying tiny signs of malevolent intent among large datasets.

2 Theory and properties of regression

A regression is a statistical method that establishes a relationship between one or more independent (explanatory) factors and a dependent variable. It is possible to determine if changes in one or more of the explanatory variables are related to changes in the dependent variable using a regression model. Based on the given predictors, these models are used to comprehend and predict the behavior of the target variable. We give a brief introduction to regression models in this section, including an outline of their main characteristics and equations. Linear regression is possibly the most commonly used regression model because of its ease of use and interpretability. It is assumed in linear regression that there is a linear relationship between the predictor variables (X) and the target variable (Y). In terms

of math, it is stated as:

$$Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \dots + \beta_n X_n + \epsilon \quad (1)$$

where:

- β_0 is the intercept term.
- $\beta_1, \beta_2, \dots, \beta_n$ are the coefficients corresponding to the predictor variables $X_1, X_2, \dots, X_n$, respectively.
- ϵ represents the error term, capturing the discrepancy between the observed and predicted values of Y .

There are various other regression models available outside linear regression, each one customized for certain situations and data distributions. Among them are:

- Polynomial Regression: In order to capture nonlinear relationships, polynomial terms are added to linear regression to create polynomial regression.
- Ridge Regression and Lasso Regression: Regularized regression techniques called Ridge Regression and Lasso Regression penalize large coefficients in order to reduce overfitting.
- Logistic Regression: When the target variable is categorical and the task is binary classification, logistic regression is utilized.

Regression analysis is a complex process, and practitioners can get important insights from their data by grasping the theory and characteristics of regression models.

2.1 L1, L2 and Elastic-net penalties

Regularization, a critical aspect of regression analysis, imposes penalties on the model coefficients to prevent them from growing excessively large. In practice, one typically uses a slightly more general form of regularization by adding a term that penalizes the size of the parameters, which will prevent overfitting [Deisenroth, Faisal, and Ong (2020)]. In this section, we delve into the distinctive properties of L1 (Lasso), L2 (Ridge), and Elastic-net penalties, elucidating their respective impacts on model complexity and predictive performance.

2.1.1 L1 regularization (Lasso)

L1 regularization, also known as Lasso (Least Absolute Shrinkage and Selection Operator), adds a penalty term proportional to the absolute value of the coefficients to the loss function. Mathematically, it can be represented as:

$$\text{Loss}_{L1} = \text{Loss}_{\text{original}} + \lambda \sum_{i=1}^n |w_i|$$

Where λ controls the strength of regularization and w_i are the coefficients of the model. L1 regularization encourages sparsity in the solution weight vector by driving some coefficients to exactly zero, effectively performing feature selection.

2.1.2 L2 regularization (ridge)

L2 regularization, or Ridge regression, adds a penalty term proportional to the square of the coefficients to the loss function. It can be represented as:

$$\text{Loss}_{L2} = \text{Loss}_{\text{original}} + \lambda \sum_{i=1}^n w_i^2$$

Similar to L1 regularization, λ controls the strength of regularization. However, unlike L1 regularization, L2 regularization does not lead to sparsity in the solution weight vector but rather shrinks the coefficients towards zero, effectively reducing their impact on the model.

2.1.3 L1-L2 regularization (elastic-net)

Elastic-net regularization combines both L1 and L2 penalties, providing a compromise between the two techniques. It adds a penalty term that is a linear combination of the L1 and L2 penalties:

$$\text{Loss}_{\text{Elastic-net}} = \text{Loss}_{\text{original}} + \lambda_1 \sum_{i=1}^n |w_i| + \lambda_2 \sum_{i=1}^n w_i^2$$

Here, λ_1 and λ_2 control the strengths of the L1 and L2 penalties, respectively. Elastic-net regularization can handle correlated features better than L1 regularization alone and still induce sparsity in the solution weight vector while also stabilizing the solution through L2 regularization.

2.2 Logistic regression

For tasks involving binary classification, supervised learning algorithms like logistic regression are employed. It simulates the likelihood that an instance, given one or more independent variables, belongs to a specific class. The probabilities produced by the logistic regression model range from 0 to 1. These probabilities are given a threshold, usually of 0.5, in order to facilitate prediction-making. Examples with predicted probability higher than the threshold are categorized as belonging to the positive class, while those with probabilities lower than or equal to the threshold are categorized as belonging to the negative class. We can generate the logistic regression model by applying the logistic function to the linear regression equation.

$$p(y = 1|x) = \frac{1}{1 + e^{-(\beta_0 + \beta_1 x_1 + \beta_2 x_2 + \dots + \beta_n x_n)}} \quad (5)$$

This equation predicts the probability that the outcome y is 1 given that the input is x . The probability that y is 0 can be calculated as $p(y = 0|x) = 1 - p(y = 1|x)$.

2.3 PCA-logistic regression

A popular dimensionality reduction method in data analysis and machine learning is principal component analysis, or PCA. Its goal is to preserve as much variance as feasible while transforming the original variables into a new collection of orthogonal variables known as principle components. In doing so, PCA maintains the key characteristics of the data while reducing its complexity. To derive PCA, let's consider a dataset X with n observations and p variables. The objective of PCA is to identify a set of orthogonal axes, expressed as linear combinations of the original variables, to maximize the variance of the data along these axes. This can be mathematically achieved by computing the covariance matrix C of X , which captures the inter-relationships among the variables. The eigenvectors of C denote the directions of maximum variance, while the corresponding eigenvalues signify the magnitude of variance explained by each eigenvector. PCA proceeds by performing eigenvalue decomposition on the covariance matrix C :

$$C = V\Lambda V^\top \quad (6)$$

where V is a matrix whose columns are the eigenvectors of C , and Λ is a diagonal matrix containing the eigenvalues. The eigenvectors are sorted in descending order based on their corresponding eigenvalues, indicating their importance in capturing the variance of the data.

3 Experiments

3.1 Internet-of-things dataset

The RT-IoT2022 dataset [Sharmila and Nagapadma (2023)] is derived from real-time IOT infrastructure, integrating a diverse range of IOT devices and complex network attacks. This dataset covers both normal and adversarial network behaviors which helps us to understand real world IOT network infrastructure. Students and researchers can utilize the RT-IoT2022 dataset to develop intrusion detection systems. Followings are some histograms of some important features useful for anomaly detection:

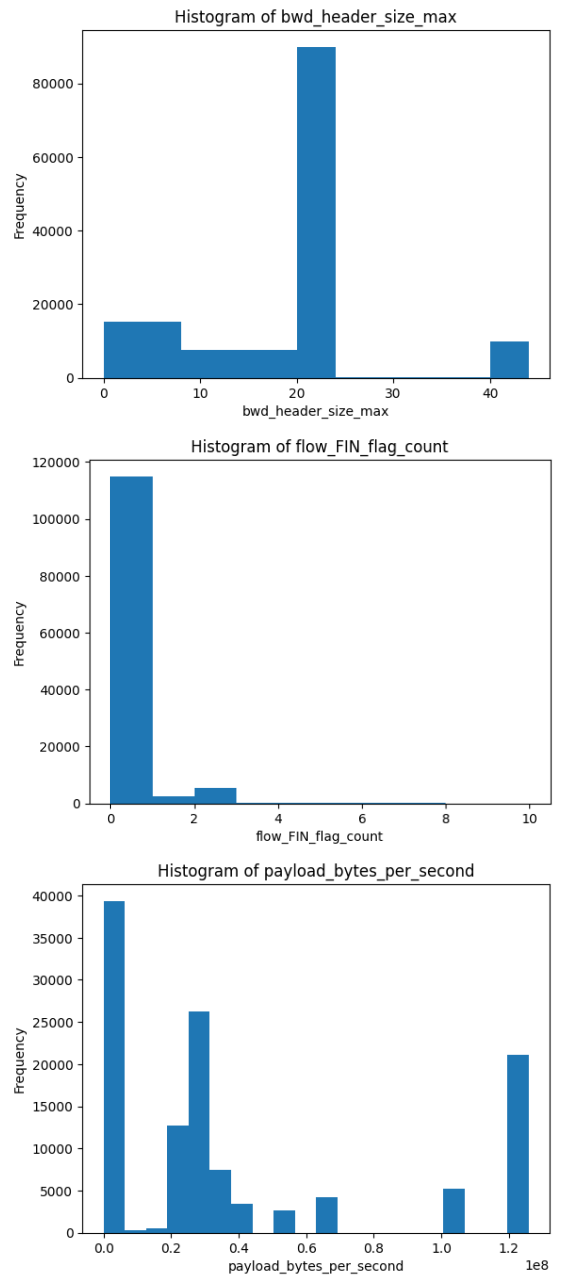


Figure 1: Histograms of features in RT-IoT 2022

3.2 Experimental setup

For preparing the data in order to train the logistic regression models I encoded the categorical data using label encoder. Since, the problem is a binary classification task, for that reason I mapped the attack-types as 1 and the non-attack-types were denoted with 0. For ensuring the robustness of my models I splitted the data into Training and Testing subsets. 70% of the data was used for training and 30% was used for testing the models. The experimentation setup was established following the method outlined in Chapter 4 of 'Hands-On Machine Learning with Scikit-Learn, Keras, and TensorFlow: Concepts, Tools, and Techniques to Build Intelligent Systems' by Aurélien Géron [Géron (2019)].

3.3 Experiment 1

For the first part of the experiment, I have run 4 logistic regression models with multiple variations of regulations. For all of the models I have specified the tolerance value for early stopping in order to avoid overfitting. For No regularization and L2 Regularization, LBFGS algorithm was used as the optimizer. For L1 regularization, I used the LIBLINEAR algorithm as solver and for Elastic-Net regularization I used the SAGA algorithm. I figured out that, L1-Ratio being 0.551 provides the best result for Elastic-Net regularization.

3.4 Experiment 2

For the second part of the experiment, I have leverage PCA method to reduce the dimension of the features by 50%. It was noticed that, we lose some information by adopting this method which was reflected in the accuracy of the model. However, the training process became way more performant. By analyzing the impact of this change, it can be inferred that, PCA method helps the models to become more generalized by reducing overfitting. From the previous experiment, it was found that L1 regularization works best on this dataset. That's why, L1 regularization was used in this experiment as well for achieving the best possible result.

3.5 Results

Five models were run, including Logistic Regression variations with different regularization techniques: No regularization, Lasso Regularization (L1), Ridge Regularization (L2), Elastic-Net Regularization (L1-L2), and PCA Decomposition with No Regularization. The accuracies are

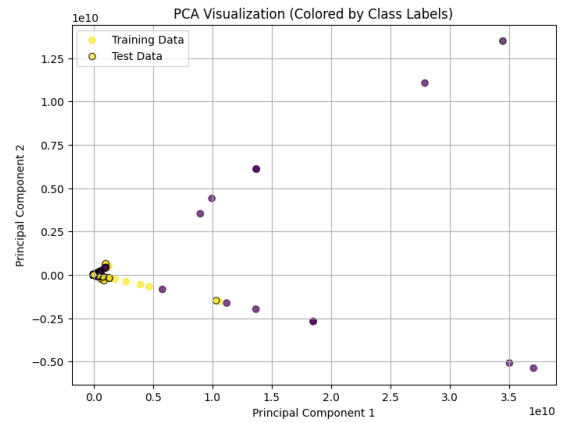


Figure 2: PCA (2 Dimensions) on RT-IoT 2022

listed below in the table.

Model	Accuracy
No regularization	95.9%
L2 regularization (ridge)	95.6%
L1 regularization (Lasso)	98.7%
L1-L2 regularization (elastic-net)	94.9%
PCA dimensionality reduction	96.1%

Table 3.5 summarises our results and will be discussed in Section 4

4 Discussion

In the study, five models using various regularization methods were compared for classification tasks. Surprisingly, the Lasso Regularization (L1) model had the highest accuracy at 98.7%, indicating its effectiveness in selecting features. On the other hand, the Elastic-Net Regularization (L1-L2) model had lower accuracy at 94.9%, indicating difficulty in balancing model complexity. The Ridge Regularization (L2) and PCA Decomposition methods showed moderate performance. These findings highlight the importance of regularization in optimizing models and stress the need to choose methods that suit the dataset and modeling goals.

5 Conclusion

To sum up, the experiment demonstrated how different regularization strategies and dimensionality reduction techniques might enhance the performance of classification models. L1 regularization, specifically through Lasso Regularization, proved to be the best performer with an amazing accuracy of 98.7%, according to thorough testing and analysis. This outcome emphasizes how important feature selection is to improving the efficiency and accuracy of the

model. It was also clear that striking the right balance between generality and complexity is necessary to achieve the best possible model performance. Elastic-Net regularization struggled to establish this equilibrium, resulting in a lesser accuracy of 94.9%, while L1 regularization excelled in feature selection. Furthermore, the application of PCA for dimensionality reduction demonstrated a trade-off between enhanced training efficiency and information loss. In general, these results highlight the importance of carefully choosing a regularization strategy that is appropriate for the features of the dataset in order to maximize model performance and generalizability in classification tasks.

References

- Deisenroth, M. P., Faisal, A. A., & Ong, C. S. (2020). *Mathematics for machine learning*. Cambridge University Press.
- Géron, A. (2019). *Hands-on machine learning with scikit-learn, keras, and tensorflow: Concepts, tools, and techniques to build intelligent systems*. O'Reilly Media, Inc.
- Sharmila, B. S., & Nagapadma, R. (2023). Quantized autoencoder (qae) intrusion detection system for anomaly detection in resource-constrained iot devices using rt-iot2022 dataset. *Journal of Cybersecurity*. Retrieved from <https://cybersecurity.springeropen.com/articles/10.1186/s42400-023-00178-5> doi: 10.1186/s42400-023-00178-5